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Paper Code: TCE 539

B.Tech. (Civil Engineering)
Odd End Semester Examination December 2025
V Semester
Building Information Modeling

Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.

Q1.

- a. Explain in detail Building Systems? CO1
- b. Explain Specification in BIM and their technical importance? CO1
- c. Explain key phases of building design process? CO1

Q2.

(2X10=20 Marks)

- a. Describe BIM and its importance in modern civil engineering? CO2
- b. Explain the differences of BIM with 3D CAD? CO2
- c. Describe BIM a part of the building design ?. CO2

Q3.

(2X10=20 Marks)

- a. Explain Mass and concept modeling in respect of Autodesk Revit? CO3
- b. Explain various users of Revit? CO3
- c. Describe the types of formats exported from Revit? CO3

Q4

(2X10=20 Marks)

- a. Explain in detail cost estimating? CO4
- b. Explain various factor of cost analysis? CO4
- c. Describe the importance of cost in construction of any civil engineering project? CO4

Q5.

(2X10=20 Marks)

- a. Explain energy modelling? CO5
- b. Explain various conflicts in in energy modelling ? CO5
- c. Discuss Interference checking in energy modelling? CO5

